SYNTHETIC / SIMULATED DATA — GENERATED BY THE RareDiseasePipeline TEST SUITE — NOT A REAL CLINICAL RESULT, NOT FOR PATIENT CARE

RareDiseasePipeline Clinical Genome Center

Synthetic Data Validation Laboratory

Pipeline QA Environment · WSL/Colab · Nextflow DSL2
Email: pipeline-qa@example-lab.org (fictional, for template purposes only)
CLIA ID# SYN-000000 (placeholder) · CAP# SYN-000000 (placeholder)

WGS FINAL REPORT (SYNTHETIC)

Patient ID	SIM-DM1-01	**Ordering Physician**	Dr. [Ordering Physician]	**Specimen**	Simulated peripheral blood
Sex	Female	**Account#**	SIM-PIPE	**Collected**	2026-07-15
Date of Birth	XX/XX/XXXX (synthetic)	**Source**	Pipeline v1.1 test dataset	**Received**	2026-07-15
Indication	Simulated: suspected myotonic disorder	**Case ID**	CS-DM1_INXXXX	**Reported**	2026-07-15

TEST RESULT: A pathogenic, heterozygous CTG-repeat expansion in **DMPK** (80 repeats; ExpansionHunter, short-read) was detected in this simulated case. Pathogenic repeat expansions in DMPK are associated with **Myotonic Dystrophy Type 1 (DM1)** [MIM: 160900], a multisystem disorder characterized by progressive muscle weakness, myotonia, cardiac conduction abnormalities, cataracts, and endocrine involvement.

PATIENT PHENOTYPE (SIMULATED)

Simulated phenotype terms used to drive prioritization: muscle weakness, myotonia, cardiac conduction abnormality.

PRIMARY FINDINGS — Variants in genes associated with patient's simulated phenotype

Confirmation Status	Gene / Locus	Condition	Chrom : Coordinates	Variant	Zygosity	Inheritance	Classification
Confirmed	DMPK	Myotonic Dystrophy Type 1	chr19:45,770,205	(CTG)n = 80 repeats	Heterozygous	Autosomal Dominant	Pathogenic

INTERPRETATION

Variant Information

This individual carries 80 CTG repeats at the DMPK locus, well above the ACMG/AMP pathogenic threshold of ≥50 repeats used by this pipeline, and far outside the normal range (≤34 repeats; 35–49 repeats is considered a premutation/intermediate allele). The repeat was sized from short-read data by ExpansionHunter and is classified Pathogenic by both the repeat-expansion rule engine and ClinVar. Two additional variants were detected in the DMPK region during annotation: a missense variant of uncertain significance (gnomAD AF 0.0009; REVEL 0.42, AlphaMissense 0.31 — both below the pipeline's pathogenic cutoffs) and a common, benign intronic variant (gnomAD AF 0.18). Neither changes the overall interpretation.

Gene Information

DMPK encodes myotonic dystrophy protein kinase. Expansion of a CTG repeat in the 3′ untranslated region of DMPK produces toxic mutant RNA that sequesters MBNL splicing factors, causing widespread mis-splicing in skeletal muscle, cardiac tissue, and the lens — the basis of DM1's multisystem presentation.

REFERENCES

Mahadevan M, et al. Science. 1992. Myotonic dystrophy mutation: an unstable CTG repeat in the 3' untranslated region of the gene.

Turner C, Hilton-Jones D. Journal of Neurology, Neurosurgery & Psychiatry. 2010. The myotonic dystrophies: diagnosis and management.

Richards S, et al. Genetics in Medicine. 2015. Standards and guidelines for the interpretation of sequence variants (PMID: 25741868).

RECOMMENDATIONS

- Clinical correlation with the patient's phenotype is recommended.
- Because this dataset is synthetic pipeline-validation data, no clinical action should be taken on it.
- Genetic counseling is recommended in real cases to discuss implications for the patient and family.
- Orthogonal confirmation (e.g. Sanger sequencing, long-read sequencing, or inverse-PCR where applicable) is recommended before clinical sign-out of any pathogenic or likely-pathogenic call.
- Periodic re-evaluation of variant classification is appropriate as knowledge and databases evolve.

TEST METHODOLOGY

This report was produced by the RareDiseasePipeline Nextflow/Colab workflow, a synthetic-data validation harness modeled on production rare-disease WGS interpretation pipelines. Variants shown here were generated by the pipeline's own per-disease test-data generator (Section 12.1 of the pipeline notebook), then passed through the same region-filtering, annotation, disease-specific-branch, prioritization, and ACMG classification steps used for real inputs: VEP and SnpEff for functional consequence, ClinVar/SnpSift for clinical significance, gnomAD for population frequency, SpliceAI/REVEL/AlphaMissense/CADD for in-silico pathogenicity prediction, InterVar for ACMG/AMP classification, ClassifyCNV and SvAnna for structural-variant interpretation, ClinGen for gene dosage sensitivity, and ExpansionHunter/TRGT for repeat-expansion sizing. Variant classification follows the ACMG/AMP 2015 sequence-variant interpretation guidelines (Richards et al. 2015).

TEST LIMITATIONS

This is a synthetic dataset generated for pipeline validation and does not represent a real patient or real sequencing data. As with any WGS-based test, full genome coverage is not achievable in repetitive or duplicated regions, and certain variant classes (e.g. deep intronic variants, some repeat expansions, complex structural rearrangements) may not be reliably detected. Structural-variant and inversion calls that are reported by only one tool in the chain are flagged as candidates and require orthogonal confirmation before they can be considered clinically actionable.

REGULATORY DISCLOSURES

LAB CONTACT

Email: pipeline-qa@example-lab.org (fictional, for template purposes only)

[Pipeline Owner Name]
Pipeline Developer / QA Lead

[Reviewer Name]
Secondary Reviewer (placeholder)

Test results reviewed and approved by: *[Reviewer Name]* — placeholder signature, synthetic report.